

Assignment 2 Report: English to Spanish Translation

Assignment Group MS BE Report

Ben Bekir Ertugrul Moritz Schlitter

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1 Results

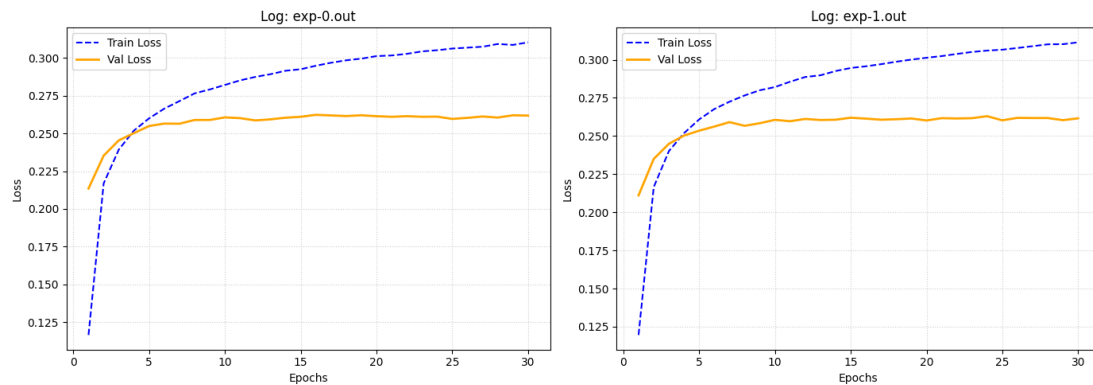


Figure 1: Training and validation accuracies for experiment 0 and 1 over the duration of the training process.

1.1 Experiment 0

Experiment 0 concerns itself with the training of a transformer model for the task of english to spanish translation. This sets a baseline performance for the following experiment, which implements a custom multi-head-attention layer.

The training process of this experiment is shown in figure 1 on the left. In epoch 1, the model achieves a training accuracy of about 0.1165. The accuracy slowly improves until the epoch 30, where it ends up at 0.3103. The validation accuracy starts at 0.21, reaches 0.2589 in epoch 8, and then stagnates. The highest validation accuracy is reached in epoch 16 with a value of 0.2623.

1.2 Experiment 1

The model for experiment 2 is built on a custom multi-head-attention layer which is used only for the encoder step inside the transformer. The rest of the architecture is identical to that of experiment 0.

Figure 1 shows the output of experiment 1 in the right corner. Being trained for 30 epochs in total, the training accuracy was increased from 0.1196 to 0.3114. The validation accuracy starts off with 0.2111 in the first epoch. Similar to experiment 0, we can see it reaching 0.2591 in epoch 7 and then stopping to improve. The final validation accuracy is 0.2616.

1.3 Experiment 2

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2 Discussion

The goal of this was to show performance differences between the usage of official and custom multi-head-attention layers.

However, we can see that there is almost no difference. The custom implementation performs slightly better for this training run, but that could be the result of a lucky random initialization for the W^q , W^k , W^v , and W^o matrices. Moreover, measuring the total time of the training process, we saw that experiment 0 took 1515.8 seconds, while experiment 1 took only 1355.2 seconds.

One of the reasons why our implementation is able to keep with the official implementation, is that we make use of similar optimizations. When using N attention heads, we do not keep N sets of matrices and do not perform our computation N times in total. Instead, we work with one regular set of matrices, as used for single-head-attention, and reshape them before applying the Softmax, so that they act as individual attention heads.

3 Conclusion

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